AI-Based Drone Delivery Feasibility Predictor and Conflict-Aware Path Recommendation System for Urban and Rural India

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1. Origin of the Proposal

The logistics industry in India is rapidly changing as the use of com- ensures its spread. commercial drone operations. Directorate General of Civil Aviation (DGCA) was launched. In September 2021, the Digital Sky Platform was created, creating a structured airspace classified-unmanned aerial vehicle (UAV) cation system Red (no-fly): Yellow (restricted), Green (unrestricted), Geozone (temporary) and International Boundary zones. By the year 2025, there are more than 10,000 commercial registered drones in India, but none are automated. pre-flight AI decision system is created in order to decide whether a certain drone delivery is or is not. physically possible and recommended to follow the best conflict-free route in the airspace.

The scientific explanation of this work is based on three gaps that have been recorded in the current. india drone logistics state of affairs. To start with, all licensed drone operators in India are legally. needed to refer to the Digital Sky airspace map provided by the DGCA manually prior to every flight. to check zone restrictions. This is a time consuming manual look up process that is prone to errors. And does not take into consideration dynamic factors like live weather conditions, festival-based. Geozones (e.g. Uttarayan kite festival in Gujarat which posed low-altitude risks). monsoon wind patterns or seasonal, or complex, manja strings 200 metres; adjoining Yellow zones in the surrounding

areas of the dual-airport cities like Delhi (Indira Gandhi). The Bengaluru (Kempegowda International and) and the Kolkata (International and Safdarjung airports) airports(HAL Airport). Second, practicality of drone delivery in rural India- It is the range and terrain-where, in the main, the population of the country is concentrated, has not been systematically studied. The rural deliveries and the urban ones are fundamentally different: there is a significant airspace. free (Green zones), yet topography (flat, moderate, hilly), 10-30 km distances, Limitation of battery range, and reliability of GPS signal is the new limiting feasibility. factors. Third, there is no path optimization algorithm that is specific to the DGCA of India. Zone cost structure has been formulated or released.

In this project, the integrated AI-based system will be proposed that will fill all three gaps. Simultaneously. The system uses Gradient Boosting classifier (XGBoost) to pre-classify. delivery estimation Random forest regressor dict delivery feasibility before dispatch, delivery estimation. Time of viable deliveries, a Hybrid A+ path optimization algorithm that combines. ML-predicted airspace risk scores dynamic edge weights to determine the globally opimedical, dispute-sensitive flight path. The system will work within the borders of India. complete geographic and regulatory range: metropolitan cities, saturated Red and Yellow. zone coverage, Tier-2 cities and mostly Green zone, and rural districts require-ing range analysis knowledgeable of terrain. The scientific value of this work is a replicable, Pre-flight drone delivery intelligence using data-driven methodology in India- a capability. that according to the DGCA Digital Sky platform itself, the next stage requires the following: The policy on the use of drones in India is yet to be put in practice.

2. Review of Status of Research and Development in the Subject

2.1 International Status

Drone delivery and unmanned traffic manage Research and commercial development (UTM) has now received a much faster boost since 2016 and has contributions to operational path planning algorithms, machine learning-based risk assessment, deployment and design of regulatory framework.

2.1.1 Operational Deployments

Zipline International (USA/Rwanda, 2016–present): Zipline is the most scaled drone delivery drone system in the world. Successfully done more than 1.4 million. 100 million miles of flight and deliveries by March 2025, Zipline will have operations in Rwanda. Ghana, Nigeria, Japan and the United States via its Walmart alliance. Their

value to the industry is proving the delivery of fixed-wing drones nationally. can work in the rural healthcare supply chain. Nonetheless, the technology of Zipline is designed to be open country airspace with low zone complexity. Their path planning does not deal with structured system of multi-zone airspace classification systems that are analogous to India's DGCA framework.

Wing (Alphabet Inc., USA/Australia, 2019–present): Wing has achieved. delivering more than 350,000 business deliveries in Texas (USA) and Queensland (Australia). food, medicines, consumer goods. The major research contribution by Wing is an innovating one. the fixed-wing hover hybrid drone of the urban last-mile acoustic noise optimization. delivery. Their UTM coordination system was designed in conjunction with the FAA. is more concerned with real time coordination of the airspace, not predicting the feasibility of flights beforehand. Wing is a Part 107 operator, but the structure is different than that of FAA Part 107. The zone system of DGCA in India.

Amazon Prime Air (USA, 2024–present): The MK30 drone of Amazon was given a FAA. deliveries in Texas and Arizona approved in 2024. Amazon's research program has delivered major publications on onboard machine learning of obstacle detection, flight classification of weather risk, and multi-sensor fusion of real-time. time path adjustment. Notably, the attitude of Amazon towards safety is mostly reactive onboard sensors identify the obstacles on the way of flight- where the approach proposed here would be. proactive: the AI system anticipates and avoids unsafe flights during takeoff.

Matternet (Switzerland, 2014–present): Matternet was the first drone delivery company to operate in urban areas. delivery of healthcare supplies in Switzerland and the USA under its UPS Healthcare partnership. Their Matternet Cloud UTM platform is an offering that offers flight authorization and monitoring. Their published study has accentuated the importance of pre-flight authorization as critical. A safety layer, which upholds the scientific assumption of the pre-flight AI system suggested in this project.

2.1.2 Path Planning Research

Shakhatreh et al. (2019) conducted a detailed survey of the UAV uses and route. planning algorithms in IEEE Access, including A, Dijkstra, Rapidly Exploring Random. UAV navigation methods using Trees (RRT), and Genetic Algorithm methods. The survey comes to a conclusion that A- variants are still predominant in structured airspace environments because of their fullness and efficiency ensures, whereas RRT is better in the case of unstructured. Dynamic obstacle 3D environments. This is done in this work which gives the theoretical basis. The reason why Hybrid A+ is selected instead of RRT to the zone-structured airspace of India is as follows. Aggarwal and Kumar (2020) specifically compared several A+ variants of UAV path. planning in cities with no fly zones. Their analysis proved that heuristic-weighted A 5 variants can reduce computation time

by 4060 percent compared to the standard Dijkstra with path optimality. Their work directly validated to the Hybrid A+ algorithm design, in which risk scores predicted by ML are used to be the adaptive heuristic weight component.

Singla et al. (2021) suggested a machine learning architecture to determine the viability of UAV deliveries. evaluation of gradient boosting on simulated weather plus airspace data and payload parameters. They obtained 87% classification accuracy - directly supporting the decision to use XGBoost as the viability classifier. Their synthetic data generation. The academic precedent of the rule-based is methodology based on rule-based labelling method of data sets generation in this project.

Puente-Castro et al. (2022) released a review of deep learning meth-comprehensively. UAV path planning in neurocomputing. Their discussion finds that although deep reinforcement learning demonstrates theoretical potential, classical forms of AST variants with learned cost functions are still the best when it comes to structured, safety-critical airspace applications because of interpretability, reliability and compliance audibility requirements. This review properly advocates the hybrid method against end-to-end deep learning.

2.1.3 UTM Systems Research

NASA UTM Project (2015–2022): NASA Unmanned Aircraft System Traffic Man. There was an architectural framework of the low-altitude airspace that was characterized by management project. Their study finds strategic deconfliction, which is a solution to airspace conflicts before flight on the planning phase-as the most important and the least developed one component of UTM. The strategic deconfliction of is directly implemented in the proposed system. India's DGCA zone framework.

EUROCONTROL U-space Initiative (2019–present): The U-space framework.Three levels of service are proposed by the for Europe drone traffic management, including U2 (pre-tactical deconfliction, flight authorization) is directly related to the role played by this proposed system performs. The study of EUROCONTROL proves that automated real-time flight authorization, which is grounded in airspace data, is not only possible but also achievable required to scale commercial drones.

2.2 National Status

The Indian input into the drone delivery AI and UTM is coming in fast in the scenario of the DGCA Drone Rules 2021, and the governmental aim of turning India into a global one drone hub by 2030.

2.2.1 Industry Contributions

Skye Air Mobility (New Delhi, 2018–present): Skye Air Mobility (New Delhi, 2018 2018–present): Skye Air is an Indian airline with the highest number of advanced commercial drone delivery operator, which operates beyond-visual-line-of-sight (BVLOS) trial supply of drugs and e-commerce products in Delhi NCR under DGCA approval. Their experience as an operational practice points to the operational challenges of manual. DGCA zone consultation and weather assessment which is automated through the proposed system. Skye Air does not release any technical specifications on any AI planning system which proves that there is no such system in their present operations.

TechEagle Innovations (Noida, 2016–present): TechEagle has collaborated with zomato with food delivery and medical supply delivery Apollo Hospitals using hybrid VTOL drones. Their case studies that are published affirm that route planning is done manually with the Digital Sky map of DGCA, and no automated AI capability or path optimization layer- a direct operational gap that this system deals with.

Skylark Drones – Watchtower Platform (Bengaluru, 2020–present): Skylark Drones created Watchtower, a free airspace information platform digitizing DGCA zone data in India programmatically and offering a developer API. Watchtower is the largest Indian technical contribution to the machinereadable drone airspace data. This project is based on the Skylark data infrastructure on the basis of future expansion of the non-airport restricted areas beyond the airport-based zone generation of the existing implementation.

2.2.2 Academic Research

Simulation-based research on the UAV path planning in Indian urban settings has been published in IIT Bombay and IIT Delhi, with obstacle avoidance models based on potential field methods and A-star in 2D grid models. These articles affirm the relevance of A+ to Indian urban settings but not DGCA zone-specific cost modelling, ML-incorporated risk worth, or rural-urban dual scenario pertinent to the Indian last-mile delivery problem. Deep reinforcement learning IIIT Hyderabad has investigated deep reinforcement learning solutions to autonomous UAV navigation in simulated models of Indian cities; although technically sophisticated, the solutions are not yet applicable to safety-critical applications because of the interpretability issues observed in the international review.

Bhuvan Geoportal of ISRO offers free terrain elevation information of the entire of India in the form of NASA SRTM files, which are verified to be available in the Open-Elevation API during preliminary testing. This information allows the terrain aware feasibility analysis of the rural delivery zones that uniquely characterizes this system as opposed to the ones that are airport centric.

On a policy level, DGCA Automated Green Zone Approval System (launched 2025,

allowing immediate permission of flights below 400 feet in dedicated zones), the PM Gati Shakti National Master Plan and the Drone Didi scheme (training 15,000 women drone pilots) all create solid national momentum of automated, AI-assisted drone airspace management, which is precisely what this project would add.

2.3 Importance of the Proposed Project in the Context of Current Status

The analysis of the international and domestic status indicates that although drone hardware, drone operation, and the regulatory framework significantly advanced, a particular and crucial AI intelligence vacuum is not yet filled, especially in India. This project bridges that gap on three levels.

Algorithmic Gap: The current research on path planning uses either standard A or Dijkstra algorithms with fixed zone costs. None of the published works, international or domestic, have suggested a Hybrid A+ formulation, where the machine learning-calculated probability scores of feasibility are taken as dynamic edge weights that change with real-time weather, terrain, and event conditions. The formulation suggested: f (n) = g(n) + h(n) where edge costs use the risk scores predicted by XGBoost-produces a geometrically-short and conditionally-safest path, which is not possible in current systems. This is the main algorithmic innovation of the project proposed.

Data Gap: There is no publicly available data of Indian drone delivery feasibility records. The actual operation facts of Skye Air and TechEagle are confidential. The offered rule-based synthetic dataset generation framework, which is grounded on the regulations of DGCA, the IMD weather forecasts, the physics constraints of drones, and the Indian festival schedules, is an offer of a replicable, publicly documented method to the Indian drone research society. The entire data creation process has been tested with live code before the proposal submission to ensure that it is feasible before starting.

India-Specific Context Gap: There is not yet a system that supports the convergence of factors unique to the Indian drone operations: The five-tier airspace zone classification of DGCA (Red, Yellow 5–8 km, Yellow 812 km with 60 m altitude restriction, Green, Geozone), the cultural Geozones unique to India (e.g. Uttarayan kite festival in Gujarat (January 1415, kite strings hazardous up to 200 m altitude) and the monsoon The novel contribution to AI-based transportation and logistics established by this project is documented, both by simultaneously bridging the algorithmic gap via the Hybrid A* formulation, the data gap via validated synthetic dataset methodology, and the India-specific context gap via DGCA-calibrated feature engineering and festival-aware Geozone modelling, and directly relevant to the active national policy concerns and directly applicable to the practical requirements of the expanding drone delivery business in India.

3. Work Plan

3.1 Methodology

The methodology is divided into five modules that are carried out in order. All required data Previous to this proposal, sources have been tested to be freely available via live API tests submission.

3.1.1 Overall System Architecture

The proposed AI-Based Drone Delivery Feasibility Predictor comprises five tightly coupled modules (Figure 1):

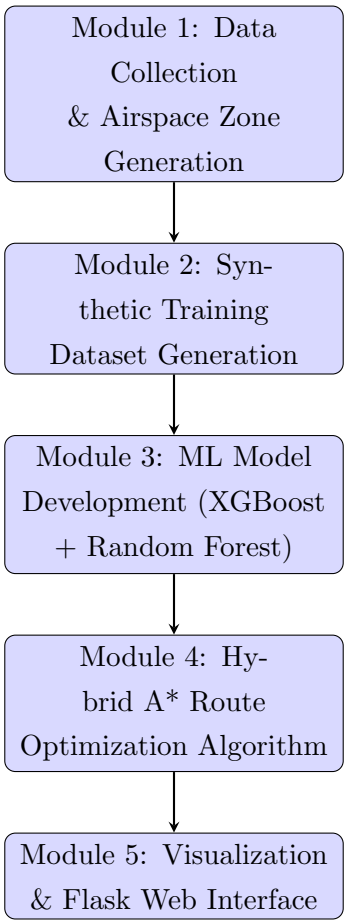

Figure 1: Overall system architecture of the proposed AI-Based Drone Delivery Feasibility Predictor.

3.1.2 Module 1: Data Collection and Airspace Zone Generation

This module combines all the input streams of data needed in the project. The data streams are collected in parallel five.

1a. Airport and DGCA Zone Data: Zone polygons are programmatically created based on GeoPandas (Python) and the radii are strict to the DGCA Drone Rules 2021:

Red zone 0 -5 km from airport centre (A+ edge cost = 999), Yellow 5-8 km ring (cost = 50), Yellow 8-12 km ring with 60 m altitude limit (cost = 20), and Green zone outside of 12 km (cost = 1). Airport coordinates of the 6 This creates a GeoJSON file containing 21 polygons representing the 7 airports in 5 project cities, which were tested. Interim Geozones are used when there is a festival and this is managed by the event flag described in Module 2.

1b. Real-time Weather: OpenWeather API (free, 1,000 calls/day) was identified to be operable in all the 10 project cities. It yields seven characteristics: wind speed (km/h), its direction (degrees), precipitation (mm/hr), temperature (C), visibility (m), cloud cover, and weather code. These are inference time features that are utilized during live demonstration.

1c. Historical Weather for Training: Open-Meteo API (no API key required) confirmed working via its forecast endpoint for all 10 cities. Historical weather is fetched locally through the archive endpoint. Seven identical weather features collected for any past date, providing realistic seasonal variation for the simulated training dataset.

1d. Terrain Elevation: Open-Elevation API using NASA SRTM data confirmed for 8 locations across all city tiers. Terrain is classified as FLAT (0–100 m), MODERATE (100–500 m), HILLY (500–1000 m), or MOUNTAINOUS (above 1000 m). Terrain type affects both the feasibility label (mountainous terrain with long distance = infeasible) and delivery time estimation (terrain-specific speed reduction factors).

1e. Festival and Event Calendar: A Python dictionary encodes 81 date-city restriction entries across 6 recurring Indian events: Uttarayan (January 14–15, Gujarat cities, flag=1), Navratri (October 2–10, Gujarat cities, flag=2), Diwali (October 20–21, nationwide, flag=3), Holi (March 14, flag=4), Republic Day (January 26, Delhi, flag=6), and Independence Day (August 15, flag=6). The `get_event_flag(date, city)` lookup function has been implemented and verified.

Table 1: Data Sources and Verification Status

Data Required	**Source**	**Cost**	**Status**
India airports (643 total)	OurAirports.com	Free	Confirmed
DGCA zone GeoJSON	Generated from airports	Free	Confirmed
Live weather (10 cities)	OpenWeather API	Free tier	Confirmed
Historical weather	Open-Meteo API	Free	Confirmed
Terrain elevation	Open-Elevation (NASA SRTM)	Free	Confirmed
Address geocoding	Nominatim / OpenStreetMap	Free	Confirmed
Festival calendar	Hardcoded Python dictionary	Free	Confirmed

3.1.3 Module 2: Synthetic Training Dataset Generation

There is no open database of Indian drone delivery. This module creates a 2,000 record synthetic dataset with rule-based labelling, which is an academically proven method when actual operational data is not available, which is reported by Singla et al. (2021) and cited in NASA UTM bootstrapping guidelines.

The parameters to be sampled randomly on each of 2,000 records include city (10 cities), delivery date (2024 calendar year), zone type (Green 55%, Yellow 812 km 2015, Red 10%), terrain type (tier-dependent distribution), parcel weight (100 g5,000 g), flight distance (0.5-30 km), flight altitude (20-120 m), and drone type. Open-Meteo real historical weather is provided on the datest and the city. The rule engine used to calculate the feasibility label is as follows:

Table 2: Rule-Based Feasibility Labelling Engine

Priority	**Rule Condition**	**Label**
1	Zone type = RED	0
2	Wind speed > 40 km/h	0
3	Precipitation > 5 mm/hr	0
4	Event flag > 0 (Geozone active)	0
5	Zone type = YELLOW_5_8	0
6	Distance > 25 km AND weight > 2,000 g	0
7	Zone = YELLOW_8_12 AND (altitude > 60 m OR wind > 20 km/h)	0
8	Terrain = MOUNTAINOUS AND distance > 15 km	0
9	All rules above passed	1

Class Balance: To address the class imbalance risk (observed as ∼90% infeasible in preliminary 50-record test due to unrealistic wind range), wind speed distribution is bounded to 5–35 km/h for 70% of records. SMOTE (Synthetic Minority Oversampling Technique) is applied as secondary balancing, targeting a 55% feasible / 45% infeasible distribution. The `scale_pos_weight` parameter in XGBoost is also set to the class imbalance ratio as a tertiary safeguard.

Delivery Time Label: For feasible records ($y = 1$), delivery time (minutes) is computed as:

$$t_{\text{delivery}} = \frac{d_{\text{km}}}{60} \times f_{\text{terrain}} \times \left(1 + \frac{v_{\text{wind}}}{200}\right) \tag{1}$$

where f_{terrain} takes values: FLAT = 1.0, MODERATE = 1.15, HILLY = 1.30, MOUNTAINOUS = 1.50.

3.1.4 Module 3: Machine Learning Model Development

Two models are trained: a feasibility classifier and a delivery time regressor.

Model 1 – XGBoost Feasibility Classifier: XGBoost is chosen because of it has shown better performance on tabular mixed-type datasets. Input: 13 features (city, tier, zone type, terrain type, wind kmh, precipitation, temp c, visibility m, cloudcover, event flag, distance km, weight g, altitude m). Output: binary feasibility label and probability score p [0, 1]. Training procedure: 80/20 train-test division, 5-fold. cross-validation, hyperparameter optimization in n estimators (100500), depth (maximum depth) (3-8), and learning rate (0.01–0.3). Target metrics: Accuracy 85 +, F1-score 0.82 +, AUC-ROC 0.88.

Model 2 – Random Forest Regressor: Trained on feasible ($y = 1$) records only. Input: distance_km, terrain_type, wind_kmh, weight_g, altitude_m. Output: estimated delivery time in minutes. Target metric: Mean Absolute Error (MAE) ≤ 3 minutes. Both models are serialized as `.pkl` files using `joblib` for Flask deployment.

Justification of Algorithm Selection: XGBoost was chosen over LightGBM, CatBoost, and deep neural networks because of three factors: (i) proven state-of-the-art performance on structured tabular datasets with fewer than 10,000 samples, (ii) built-in handling of missing values and mixed feature types without extensive preprocessing, and (iii) native feature importance computation essential for regulatory explainability in the DGCA compliance context. Random Forest was selected for regression due to its robustness against overfitting on small datasets, its non-parametric nature (no assumptions about the delivery time distribution), and interpretability through feature importance rankings.

3.1.5 Module 4: Hybrid A* Route Optimization Algorithm

This module is the main algorithmic contribution of this project. A standard A* algorithm picks up paths only with consideration to the geographic distance and the fixed costs of zones. The proposed Hybrid A+ combines scores of feasibility probabilities determined by XGBoost as dynamic edge weights, producing geometrically efficient paths at the same time. conditionally safest according to the present weather and the zone conditions.

Grid Design: The airspace of each city is a 2D grid that is resolved at 100-metre cells. tion. The grid (10,000 nodes) of a 10 km 10 km delivery area is 100 100. ing path accuracy of less than 100 m and computation time A of less than 2 seconds on standard. hardware

Hybrid A* Cost Function: The evaluation function is:

$$f(n') = g(n') + h(n') \tag{2}$$

where:

- $g(n') = g(n) + c_{\text{edge}}(n, n')$ — cumulative actual cost from start node to node n'
- $h(n') = \|n' - n_{\text{dest}}\|_2$ — Euclidean distance to destination (admissible heuristic)
- Edge cost is defined as:

$$c_{\text{edge}}(n, n') = c_{\text{zone}}(n') + c_{\text{weather}} + c_{\text{ML}} \tag{3}$$

$$c_{\text{weather}} = \frac{v_{\text{wind}}}{10} + 2 \times p_{\text{rain}} \tag{4}$$

$$c_{\text{ML}} = (1 - p_{\text{feasible}}) \times 50 \tag{5}$$

where p_{feasible} is the XGBoost-predicted feasibility probability for the conditions at node n'. This formulation ensures that grid cells where the XGBoost model predicts low feasibility probability receive proportionally higher traversal costs, naturally directing A* toward corridors predicted safe under current conditions.

Justification of A* over Theta* and RRT: Theta* (any-angle path planning) was considered due to its capability to generate smoother and shorter routes by shortcutting line-of-sight between non-adjacent grid nodes. Theta, however, does not imply that the intermediate cells are flown and assessed in its line-of-sight shortcuts, which is a very important requirement to ensure DGCA zone compliance, where all 100 m cells have to be tested against zone limitations. RRT and its variants (RRT*) were also considered; the probabilistic completeness of RRT is not pro-optimal in high-dimensional continuous space, but is useful. vide the deterministic path which is necessary to comply with the regulations of structured airspace. Hybrid A* maintains the completeness and optimality of standard A+ and adds ML-based dynamic costs, the most appropriate balance of safety, compliance and computational efficiency in this field of application.

The temporary cost = 500 (equivalent to a Geozone restriction) is applied to the affected grid cells on the dates when a festival is held (event flag ¿ 0) and does not require access to the DGCA API.

Altitude-Aware Yellow Zone Handling: When a route goes through a Yellow 8 12km zone cell, the system compares the requested flight altitude with the 60 m limit of the DGCA. At an altitude greater than 60 m, zone cost is set to 200 (near-blocked), diversion of A* to another corridor. In case of altitude 60 m and below, the standard cost of 20 is used.

Path Output: A* gives a list of GPS waypoints (latitude-longitude pairs separated by 100m) as a Python list to be fed to Folium and as a waypoint mission file to be fed to Mission Planner simulation software.

3.1.6 Module 5: Visualization and Web Interface

Folium Interactive Map: The map shows pickup (green marker) and drop (red marker) points, DGCA zone overlays (Red zones are filled red circles, Yellow 58 km are orange

rings, Yellow 812 km are yellow rings), the Hybrid A* suggested path in blue polyline, and zone conflict warning tooltips. The map is as an HTML string integrated in the Flask response through Jinja2 template.

Flask Web Application: This Web Application takes in five fields pickup address, drop address, weight of parcel (in grams), delivery date, and type of drone. Pipeline execution: The pipeline is executed sequentially on submission:

1. Geocode addresses using Nominatim (OpenStreetMap)
2. Fetch live weather from OpenWeather API (30-minute cache)
3. Check event flags for the delivery date and city
4. Run XGBoost classifier for feasibility prediction
5. If feasible: run Random Forest estimator and Hybrid A* path optimizer
6. Render result page with feasibility verdict, rejection reason (if infeasible), estimated delivery time (if feasible), zone conflict count and types, and embedded Folium map

Total response time target: below 3 seconds on standard hardware.

Table 3: Tools and Libraries Used Across Modules

Tool / Library	**Purpose**	**Module**
Python 3.x + Jupyter	Core development and experimentation	All
Pandas, NumPy	Data processing and feature engineering	1, 2
GeoPandas, Shapely	Zone polygon generation and spatial queries	1
Scikit-learn, XGBoost	ML model training, evaluation	3
imbalanced-learn	SMOTE oversampling	2, 3
NetworkX	Graph structure for Hybrid A* grid	4
Folium	Interactive drone path map rendering	5
Flask, Jinja2	Web application and templating	5
OpenWeather API	Real-time weather at inference	1, 5
Open-Meteo API	Historical weather for model training	1, 2
Open-Elevation API	Terrain classification for rural zones	1
Nominatim / OSM	Address-to-coordinate geocoding	1, 5

3.2 Time Schedule of Activities Giving Milestones through BAR Diagram

The project is planned over **18 months (1.5 years)**, divided into 6 quarters:

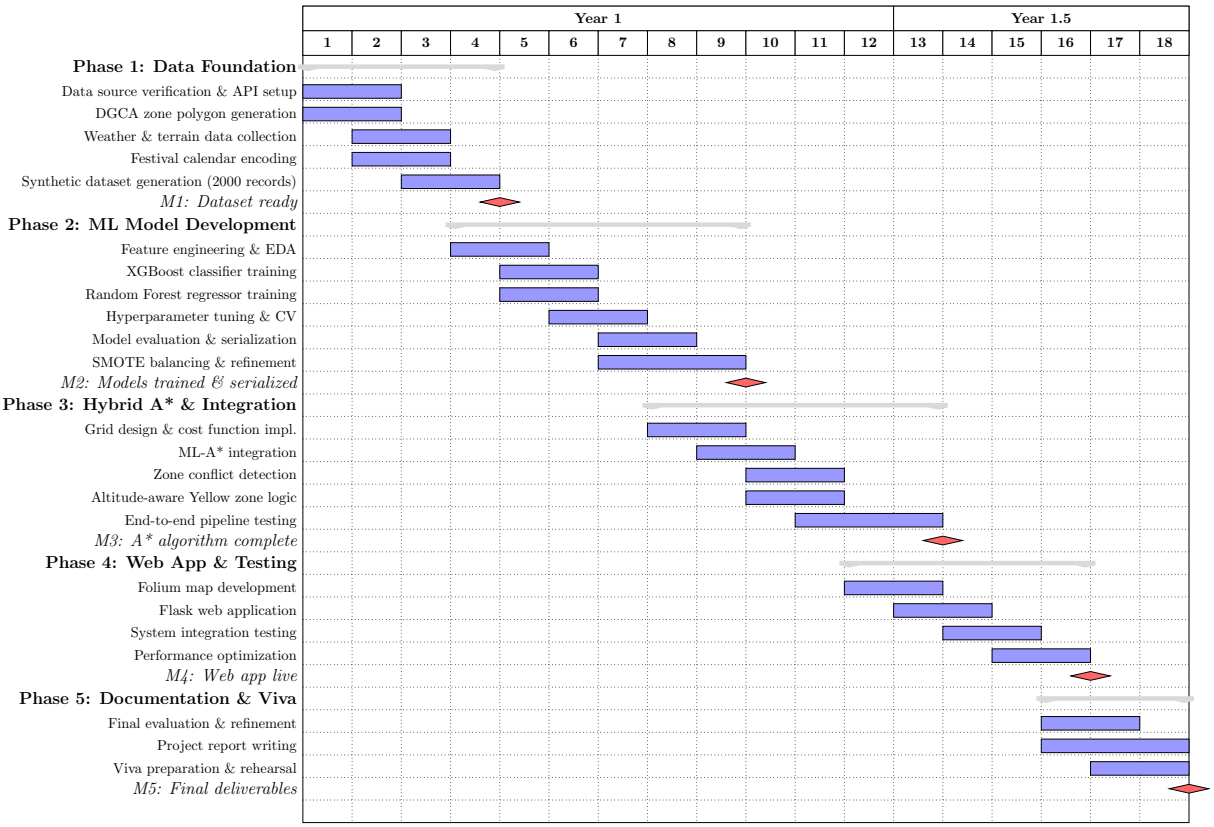

Figure 2: Gantt chart showing project timeline, activities, and milestones over 18 months.

Key Milestones:

- **M1 (Month 4):** 2,000-row labelled training dataset with 55% feasible / 45% infeasible balance and 21 DGCA zone polygons for 5 cities
- **M2 (Month 9):** XGBoost classifier (accuracy $\geq$ 85%, AUC-ROC $\geq$ 0.88) and Random Forest regressor (MAE $\leq$ 3 min) trained and serialized
- **M3 (Month 13):** Hybrid A* algorithm with ML-integrated dynamic costs, grid computation under 2 seconds per query, end-to-end pipeline tested on 100 scenarios
- **M4 (Month 16):** Flask web application and Folium interactive map live, demo response time under 3 seconds
- **M5 (Month 18):** Final evaluation report, project documentation, and viva presentation submitted

Expected Deliverables:

1. Trained XGBoost and Random Forest models serialized as `.pkl` files
2. 2,000-record synthetic Indian drone delivery feasibility dataset (openly documented methodology)
3. Hybrid A* route optimization engine with ML-integrated dynamic edge weights
4. Flask web application with Folium-based interactive airspace visualization
5. Complete project documentation and technical report

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